University of Hertfordshire UH
School of Physics, Engineering and Computer Science

Coursework – Group Report Briefing Sheet (2026/27 Academic Year)

Weighting %:	70% of Coursework	**Target for returning coursework:**	4 weeks from the date of the submission
Authorship:	Group	**Number of hours you are expected to work on this assignment:**	15 hours

This Assignment assesses the following Learning Outcomes:

I. Have a deep and systematic understanding of the fundamental concepts underlying procedural programming.
II. Have a deep and systematic understanding of the fundamental concepts and facilities offered by a modern object-oriented programming language.
III. Be able to write procedural programs as solutions to complex and clearly specified problems.

Project Title: Interactive Kotlin Command-Line Utility

Objective: Develop a Kotlin-based command-line application that processes user inputs to perform specific calculations and conversions. The application will help students apply procedural programming concepts and implement control flow, data handling, and user input validation using Kotlin.

Requirements:

1. **Command Handling:**
   - Implement a single function to interpret and execute commands interactively.
   - Commands to implement (for processing every command a separate function should be implemented):
     - **sum**: Calculates $\left(1+\frac{1}{2}+\frac{1}{3}+\ \ldots\ +\frac{1}{n}\right)$, where *n* is a natural number.
     - **gcd**: Calculates the greatest common divisor (GCD) of two given positive integers *a* and *b*.
     - **fibonacci**: Calculates the sum of the first *n* Fibonacci numbers, where *n* is a natural number.
2. **Input Validation:**
   - Validate all user inputs to ensure they are suitable for the requested calculation. Inputs should be numeric and within the required range; for example, reject negative numbers where positive values are required, decimal values where integers are expected, and empty or non-numeric inputs. Display a clear error message when invalid input is entered.
   - Provide meaningful error messages if inputs are invalid or out of the expected range.
3. **User Interaction:**
   - The application must interactively request and retrieve input from the console.
   - Display results or error messages back to the user in a clear and understandable format.
4. **Documentation and Reporting:**
   - Document the project design and implementation in a detailed report.
   - Include roles and contributions of each team member in implementation and documentation.
   - Provide the complete Kotlin code and explain the logic and functionality.
   - Create a flowchart diagram to visually represent the algorithm.
   - Include screenshots of the code within the IntelliJ IDEA environment to demonstrate the development and testing stages. Screenshot of code should be supplemented with screenshots of program execution and screenshots demonstrating validation. Also, if possible, the student ID number should be included in the output screenshots for verification.

The project will be completed in small teams. Team members have already been assigned, and you can find your team and teammates in the **People > Groups** section of Canvas. Each team should hold a meeting to discuss and divide the tasks evenly among members.

The report should be approximately 5 pages or 1500 words (font size 11, Arial) to be structured to address specific aspects of the assignment. In your group report, please include the names of your team members with their student IDs.

Submission Requirements:
The assignment must be submitted online via StudyNet. Submit your report as a .pdf file.

Marks awarded for:

Criteria	Excellent	Good	Satisfactory	Poor	%
Functionality	All commands are implemented correctly and robustly.	Minor issues in one command, but overall functional.	Major issues in one or more commands, but runs.	Commands not functioning as required.	40
Code Quality	Code is well-organised, and optimally written.	Code is mostly clean with some inefficiencies.	Code has redundancy, needs refactoring.	Code is messy, difficult to read, or inefficient.	10
Error Handling	Comprehensive error checks and clear messages.	Good error handling with few missing cases.	Basic error handling; some cases not covered.	Poor or no error handling; unclear messages.	10
Documentation	Comprehensive and clear documentation and diagrams.	Well-documented but lacks detail in some areas.	Adequate documentation; some areas poorly explained.	Insufficient documentation; many areas not covered.	20
Team Contribution	Detailed and clear explanation of team roles.	Good description of roles with minor omissions.	Some description of roles but lacking clarity.	Roles not clearly defined or missing.	10
Presentation	Code and reports are exceptionally well-presented.	Well-presented with minor errors in format.	Presentation is acceptable but lacks polish.	Poor presentation, difficult to follow.	10
Overall Comments and marks	70% of this group assignment will contribute towards your coursework grade.				**100**

Type of Feedback to be given for this assignment:

Formative verbal feedback can be given during the drop-in sessions as per the module delivery plan. The assignment will be marked according to the above marking scheme. You should address each of the marking components within your assignment.
You can ask your questions related to the assessment activities in the discussion section in StudyNet. These questions and the answers will help you and other students to reflect on the activities you will be undertaking as part of the assessed work for the module. Self-assess your work as you progress through the module which will help you understand the subject better. Feedback is not just the marks and the commentary at the end of the assessment – it is also the regular advice about your work as you undertake the practical activities for the rest of assignments. If you fail to undertake the practical activities and you fail to engage with the class and with the instructors, you will disadvantage yourself.

Additional information:

- The students' reports are checked with Turnitin software for plagiarism purposes.
- The use of generative AI such as ChatGPT is strictly prohibited in completing this assignment.
- Regulations governing assessment offences including Plagiarism and Collusion are available from https://www.herts.ac.uk/__data/assets/pdf_file/0007/237625/AS14-Apx3-Academic-Misconduct.pdf (UPR AS14).
- Guidance on avoiding plagiarism can be found here: https://herts.instructure.com/courses/61421/pages/referencing-avoiding-plagiarism?module_item_id=779436
- For post**graduate modules**:
    - a score of 50% or above represents a pass performance at honors' level.
    - late submission of any item of coursework for each day or part thereof (or for hard copy submission only, working day or part thereof) for up to five days after the published deadline, coursework relating to modules at Level 7 submitted late (including deferred coursework, but with the exception of referred coursework), will have the numeric grade reduced by 10 grade points until or unless the numeric grade reaches or is 50. Where the numeric grade awarded for the assessment is less than 50, no lateness penalty will be applied.